EEX5362 – Performance Modelling

Performance Modeling of a Public Transportation Network

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1. High-Level Problem Statement

1.1 Background

Public transportation plays a critical role in ensuring mobility, accessibility, and sustainability in urban environments. In cities like Colombo, buses are the primary mode of public transit, serving thousands of commuters daily. However, inefficiencies such as irregular schedules, long waiting times, bottlenecks at key stops, and overcrowding significantly affect passenger satisfaction and overall system performance.

Traditional manual scheduling methods fail to account for real-time dynamics like variable boarding times, traffic congestion, and fluctuating passenger demand. Therefore, a performance modeling approach is essential to quantitatively evaluate how the network behaves under different conditions and to design strategies that enhance efficiency, reliability, and scalability.

1.2 Problem Definition

The project aims to model, simulate, and analyze the performance of Colombo's urban bus network by focusing on selected high-density routes. Using discrete-event simulation, this study seeks to capture operational parameters (e.g., dwell times, headways, passenger load variations) and evaluate how these influence overall system performance.

By replicating real-world bus movement and passenger behavior, the model will provide a data-driven framework to identify bottlenecks, optimize scheduling, and propose improvements to resource allocation.

1.3 Purpose of the Study

- To simulate daily operations of the public transportation system.
- To identify performance bottlenecks (such as delays, idle times, or route congestion).
- To evaluate the impact of operational parameters (headway, dwell time, boarding rates) on throughput and reliability.
- To suggest strategies for performance improvement (better scheduling, fleet adjustment, and stop-time optimization).

1.4 Significance

- Transport authorities in identifying inefficiencies and planning new schedules.
- Bus operators in managing fleets dynamically based on demand.
- Passengers through reduced waiting time and improved reliability.
- Researchers and students as a foundation for future smart city transport modeling.

2. System Description

2.1 Selected System

System Name: Colombo Urban Bus Network

System Type: Public transportation system

Scope: Modeling and simulation of real-world bus operations covering routes 103 (Narahenpita–Fort), 144 (Rajagiriya–Fort), and 171 (Koswatta–Fort).

2.2 System Complexity

The chosen transportation network is complex and dynamic, involving interactions between:

- Multiple buses operating on overlapping routes
- Numerous passenger boarding/alighting events
- Varying traffic conditions and stop dwell times
- Time-dependent variations (peak vs off-peak hours)

The complexity lies in capturing the interdependency of these variables — how passenger load affects dwell time, how headway changes influence congestion, and how schedule delays cascade across the system.

2.3 Key Components

Component	Description
Bus Units	Each bus has unique ID, route number, and schedule.
Bus Stops	Defined set of stops along each route (e.g., Borella, Maradana, Fort).
Passengers	Defined set of stops along each route (e.g., Borella, Maradana, Fort).
Schedules	Defined frequency (headway), travel time, and stop sequence per route.
Simulation Clock	Tracks time events such as arrivals, departures, and delays.

2.4 System Operation Flow

- Bus begins at route origin (e.g., Koswatta).
- Travels through each stop based on the defined schedule.
- Passengers board and alight according to stop demand.
- Dwell time is computed dynamically (depending on number of passengers).
- Headway (gap between buses) is monitored to identify irregular spacing.
- Data such as arrival time, departure time, and occupancy are logged continuously.

3. *Dataset Details*

3.1 Dataset Description

The dataset used for this project is synthetic but realistic, designed to represent daily operations across selected routes. It captures temporal, spatial, and operational parameters that reflect real-world variability.

Attribute	Description	Example
bus_id	Unique bus identifier	103A_01
route_no	Bus route number	171
direction	Inbound / Outbound	Inbound
stop_name	Stop where event occurs	Borella
arrival_time	Arrival timestamp	08:15:23
departure_time	Departure timestamp	08:16:10
boarded	Number of passengers boarding	12
alighted	Number of passengers leaving	5
occupancy_after	Occupancy after the stop	38
dwell_time_s	Dwell time in seconds	47
headway_s	Time gap between consecutive buses on same route	300

timepoint	Whether the stop is a timing checkpoint (1/0)	1
is_peak	1 if within peak hours	0

3.2 Dataset Use

- Input for Simulation: Used to parameterize and calibrate SimPy model.
- Validation: Cross-check modeled metrics with empirical (field) observations.
- Output Analysis: Used to calculate average wait time, throughput, and utilization.

3.3 Dataset Format

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4. *Performance Objectives*

Performance modeling focuses on quantifiable characteristics that describe system efficiency, capacity, and responsiveness.

3.3 Primary Objectives

Objective	Description	Metric
Minimize Passenger Waiting Time	Reduce average wait time at stops by optimizing headway	Average wait time (min)
Maximize Throughput	Increase number of passengers transported per route per hour	Passengers/hour
Identify Bottlenecks	Detect stops or intervals causing congestion	Delay duration (s)
Optimize Resource Allocation	Determine minimum buses required to meet demand	Utilization rate (%)
Evaluate Schedule Adherence	Measure deviation from planned timetable	Delay index
Enhance Scalability	Evaluate performance under increased passenger loads	Performance degradation %

3.4 Secondary Objectives

- Compare performance during peak vs off-peak hours.
- Visualize occupancy and throughput trends.
- Predict future performance under “what-if” scenarios (e.g., route expansion, more buses).

5. *Performance Metrics*

Metric	Unit	Purpose
Throughput	Passengers/hour	Measure of efficiency
Queue Length	Number of passengers	Indicator of demand/supply gap
Dwell Time	Seconds	Time lost at each stop
Utilization	%	Percentage of bus capacity used
Latency	Seconds	Time between successive arrivals
Scalability Index	%	Change in performance under higher loads

6. *Expected Outcomes*

- A quantitative model of bus system dynamics using discrete-event simulation.
- Graphs and visualizations (queue length vs time, average waiting time by stop).
- Identification of bottlenecks and delayed segments.
- Recommendations for optimal bus scheduling and fleet distribution.
- Insights into how system performance changes under different demand scenarios.

7. *Tools and Technologies*

Tool	Purpose
Python (SimPy)	Discrete-event simulation
Pandas	Data manipulation
Matplotlib / Seaborn	Visualization
GitHub	Version control and repository sharing
MS Excel / CSV	Dataset preparation

8. *Repository Details*

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README.md includes:

- Project overview
- Dataset explanation
- Instructions to run simulation
- Expected output summary

9. *Conclusion*

This project lays the foundation for a data-driven, simulation-based performance analysis of public bus systems. By modeling passenger behavior, traffic conditions, and scheduling patterns, it provides actionable insights for optimizing routes, minimizing waiting times, and enhancing service reliability.

The proposed simulation framework is flexible, scalable, and adaptable for other urban transport systems, making it a valuable contribution toward smart mobility planning.